

Small signal stability analysis of power systems with DFIG based wind power penetration



Bhinal Mehta^a, Praghnesha Bhatt^{a,*}, Vivek Pandya^b

^a Department of Electrical Engineering, C.S. Patel Institute of Technology, CHARUSAT, Changa, Gujarat, India

^b Department of Electrical Engineering, School of Technology, PDPU, Gandhinagar, Gujarat, India

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ABSTRACT

With the increasing penetration of wind power generation into the power system, it is required to comprehensively analyze its impact on power system stability. The present paper analyzes the impact of wind power penetration by doubly fed induction generator (DFIG) on power system oscillations for two-area interconnected power system. The aspects of inter-area oscillations which may affect the operation and behaviour of the power systems are analyzed with and without the wind power penetration. Eigenvalue analysis is carried out to investigate the small signal behaviour of the test system and the participation factors have been determined to identify the participation of the states in the variation of different mode shapes. The penetration of DFIG in a test system results in an oscillatory instability, which can be stabilized with the coordinated operation of automatic voltage regulator (AVR) and power system stabilizer (PSS) equipped on synchronous generators. Also, the variations in oscillatory modes are presented to observe the damping performance of the test system at different wind power penetration level.

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1. Introduction

The electricity industry worldwide is turning increasingly to renewable sources of energy to generate electricity. Wind is the fastest growing and the most widely utilized emerging renewable energy technology for power generation at present, with a total of approximately 250 GW installed worldwide up to 2012 [1].

The wind turbine generators (WTGs) are divided into two basic categories: fixed speed and variable speed. A fixed-speed WTG generally uses a squirrel-cage induction generator (SCIG) to convert the mechanical energy from the wind turbine into electrical energy. DFIG and direct drive synchronous generator (DDSG) are popular types of variable speed WTGs. Variable-speed WTGs can offer increased efficiency in capturing the energy from wind over a wider range of wind speeds, along with better power quality and with the ability to regulate the power factor, by either consuming or producing reactive power. In the DFIG, the rotor is connected to the power system through the back-to-back ac/dc/ac converter, while the stator is connected directly to the power system. The control scheme of DFIG decouples the rotational speed of rotor from the grid frequency [2,3].

Modelling of DFIG for stability studies has lead to various models developed using different approaches presented in [4–8]. The influence of load increase, the length of transmission network

interface and the different penetration levels of a constant speed wind turbine generator on power system oscillations is studied in [9] with the consideration of SCIG.

In [10], supplementary control strategy is designed for the DFIG power converters to mitigate the impact of reduced inertia due to significant DFIG penetration in a large power system. Small signal behaviour of DFIG in power factor control mode and voltage control mode were extensively analyzed in [11]. Improved controller tuning is proposed to damped inter area mode oscillations. The impacts of DFIGs on the electromechanical modes are demonstrated to be highly dependent on their control strategies. The authors in [12,13] have reported the likelihood of the contribution of DFIGs to the system damping with application of appropriate coordinated tuning of controllers by evolutionary techniques. In [14], an approach of sensitivity analysis of electromechanical modes to the inertia of the generators is introduced and the results have shown both detrimental and beneficial impacts of increased DFIG penetration into the power system.

The dynamic behaviour of the DFIG has been investigated by various authors. The majority of these studies are to show the impact of DFIG on power system dynamics, the merits of decoupled control and maximum power tracking and the response to grid disturbances, the fault ride through behaviour, the control methods to make the DFIG behave like a synchronous generator [15–20]. The advanced control capabilities of DFIG are used in [21] to enhance network damping via an auxiliary power system stabilizer loop.

* Corresponding author. Mobile: +91 9427339896.

E-mail address: praghneshbhatt.ee@ecchanga.ac.in (P. Bhatt).

The literature survey shows that the small signal stability analysis with the integration of wind power generation has been addressed for single machine infinite bus system (SMIB) whereas the less attention has been paid for multi machine system which are equipped with AVR and PSS on synchronous machines. The modes of response introduced by DFIG in interconnected power systems, as well as the effect of increased penetration of DFIG on inter area oscillations considering two-area power system have been investigated in the present paper. The influence on modes with the displacement of synchronous generators equipped with or without AVR and PSS are shown. Further, the effects of placement of DFIG at different locations and stepwise penetration of wind power on small signal stability are analyzed in the present work.

The paper is organized in six sections. Section 2 presents the characteristics and modelling concepts associated with DFIGs. The small signal stability and eigenvalue analysis has been discussed in Section 3. Section 4 details the approach developed to analyze the impact of increased penetration of DFIGs on small signal and transient stability along with case description. Simulation and results with different cases are presented and discussed in Section 5 followed by the conclusion in Section 6.

2. Characteristic and modelling of DFIG

2.1. Electrical dynamics

The schematic of DFIG interfaced with electrical grid is shown in Fig. 1. The stator and rotor flux dynamics of induction machine are assumed to be fast in comparison with grid dynamics and the converter controllers decouple the generator rotational speed from the grid. As a result, steady-state electrical equations of DFIG can be represented by (1)–(4) [22–25].

$$v_{ds} = -R_s i_{ds} + ((X_s + X_m) i_{qs} + X_m i_{qr}) \quad (1)$$

$$v_{qs} = -R_s i_{qs} - ((X_s + X_m) i_{ds} + X_m i_{dr}) \quad (2)$$

$$v_{dr} = -R_r i_{dr} + (1 - \omega_m)((X_r + X_m) i_{qr} + X_m i_{qs}) \quad (3)$$

$$v_{qr} = -R_r i_{qr} - (1 - \omega_m)((X_r + X_m) i_{dr} + X_m i_{ds}) \quad (4)$$

where v_{ds} and v_{qs} are d - and q -axis stator voltages, respectively, v_{dr} and v_{qr} are d - and q -axis rotor voltages, respectively, i_{ds} and i_{qs} are d - and q -axis stator currents, respectively, i_{dr} and i_{qr} are d - and q -axis rotor currents respectively, R_s is stator resistance, R_r is rotor resistance, X_s is stator reactance, X_m is magnetization reactance, X_r is rotor reactance and ω_m is the rotor speed.

2.2. Power dynamics

The power injected in the grid depends on the operating mode of the converter and it is also a function of stator and rotor currents. Hence, the converter powers on the grid side and rotor side are represented by (5)–(6) and (7)–(8), respectively.

The converter powers on grid side:

$$P_g = v_{dg} i_{dg} + v_{qg} i_{qg} \quad (5)$$

$$Q_g = v_{qg} i_{dg} - v_{dg} i_{qg} \quad (6)$$

The converter powers on rotor side:

$$P_r = v_{dr} i_{dr} + v_{qr} i_{qr} \quad (7)$$

$$Q_r = v_{qr} i_{dr} - v_{dr} i_{qr} \quad (8)$$

where v_{dg} and v_{qg} are d - and q -axis voltages of the grid side converter respectively, i_{dg} and i_{qg} are d - and q -axis current of the grid side converter, respectively.

2.3. Wind turbine dynamics

In stability studies of DFIG wind turbine, the model of drive train is important. It is assumed that the converter controls are able to filter shaft dynamics and therefore the generator motion equations, which represent mechanical dynamics of wind turbine, are as follows:

$$\dot{\omega}_m = (T_m - T_e)/2H_m \quad (9)$$

$$T_e = X_m(i_{qr} i_{ds} - i_{dr} i_{qs}) \quad (10)$$

where T_e is electromagnetic torque and is approximated as follows:

$$T_e \approx -\frac{X_m V_{igr}}{\omega_b (X_s + X_m)} \quad (11)$$

where ω_b is the system frequency rate in rad/s, H_m is inertia constants in kW s/kV A. The mechanical torque T_m which is the power input of the wind turbine is as follows:

$$T_m = \frac{P_w}{\omega_m} \quad (12)$$

where P_w the mechanical power extracted from the wind and is as follows:

$$P_w = \frac{\rho}{2} C_p(\lambda, \theta_p) A_r v_w^3 \quad (13)$$

where ρ is the air density, v_w is the wind speed, θ_p is the pitch angle, A_r is the area swept by the rotor and λ is the blade tip speed ratio.

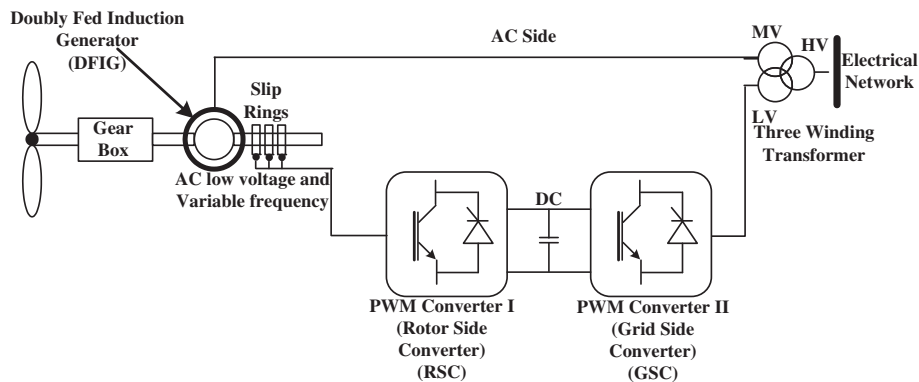


Fig. 1. Schematic of DFIG integrated with electrical grid.

The performance co-efficient or the power co-efficient c_p is as follows:

$$c_p = 0.22 \left(\frac{116}{\lambda_i} - 0.4\theta_p - 5 \right) e^{-\frac{12.5}{\lambda_i}} \quad (14)$$

$$\text{With } \frac{1}{\lambda_i} = \frac{1}{\lambda + 0.08\theta_p} - \frac{0.035}{\theta_p^3 + 1} \quad (15)$$

$c_p(\lambda, \theta_p)$ has a maximum $c_p^{\max}$ for a particular tip speed ratio λ_{opt} and pitch angle $\theta_p = 0$. The aim for variable wind turbine at wind speeds lower than rated value is to adjust the rotor speed at varying wind speeds; therefore, λ and c_p are always maintained at the optimal and maximum value, respectively. The speed control of the DFIG is achieved by driving the generator speed along the optimum power–speed characteristic curve (see Fig. 5), which corresponds to the maximum energy capture from the wind. In this curve, when generator speed is less than the low limit or higher than the rated value, the reference speed is set to the minimal value or rated value, respectively. When generator speed is between the lower limit and the rated value, the rotor speed reference can be obtained by substituting λ into (13) as follows:

$$\lambda = \frac{v_t}{v_w} = \eta_{GB} \frac{2R\omega_{wr}}{p v_w} \quad (16)$$

where v_t is the blade tip speed, v_w the wind speed upstream the rotor, η_{GB} is the gear box ratio, p is the number of poles of the induction generator and R is the rotor radius.

2.4. Pitch angle control

The pitch angle of the blade is controlled to optimize the power extraction of wind turbine as well as to prevent overrated power production in strong wind. The pitch servo is modelled as follows:

$$\dot{\theta}_p = (K_p \varnothing(\omega_m - \omega_{ref}) - \theta_p) / T_p \quad (17)$$

where $\varnothing$ is a function which allows varying the pitch angle set point only when the difference $(\omega_m - \omega_{ref})$ exceeds a predefined value $\pm \Delta\omega$.

For the sake of simplicity, the reference of the pitch angle $\theta_p = 0$ when wind speed is below rated value. When wind speed is higher than rated value, the power limitation is active by adjusting the pitch angle using the pitch-control scheme shown in Fig. 2.

2.5. Converter dynamics

The ac–dc–ac converter model in DFIG system comprises of two pulse width modulation invertors connected back to back via dc link as shown in Fig. 1. The rotor side converter (RSC) and the grid side converter (GSC) act as a controlled voltage source and controlled current source, respectively. The RSC injects an ac voltage at slip frequency to the DFIG rotor, whereas the GSC injects an ac current at grid frequency to the network and maintains the dc-link voltage constant.

Converter dynamics are fast with respect to the electromechanical transients, hence, it is modelled as an ideal current source, where i_{qr} and i_{dr} are state variables and are used for the rotor speed

control and the voltage control, respectively, which are depicted in Figs. 3 and 4.

Differential equations for the converter currents are as follows:

$$\dot{i}_{qr} = \left(-\frac{X_s + X_m}{X_m V} P_w^*(\omega_m) / \omega_m - i_{qr} \right) \frac{1}{T_e} \quad (18)$$

$$\dot{i}_{dr} = K_V(V - V_{ref}) - V/X_m - i_{dr} \quad (19)$$

where $P_w^*(\omega_m)$ is the power–speed characteristic which roughly optimizes the wind energy capture and is calculated using the actual rotor speed value (see Fig. 5).

It is assumed that $P_w^* = 0$ if $\omega_m < 0.5$ pu and that $P_w^* = 1$ pu if $\omega_m > 1$ pu. Thus, the rotor speed control only has effect for sub-synchronous speeds. Both the speed and voltage controls undergo anti-windup limiters in order to avoid converter over currents.

In this paper, it is assumed that the ac–dc–ac converter is made of lossless components, and the switching dynamics are not considered. The analysis is done for open loop, i.e. the rotor voltage remains constant at its initial value. Closed-loop controllers are not included because the eigenvalues displacement they introduce depends very much on the control scheme and gains, so that the conclusions regarding the small-signal behaviour will not be a true reflection of the DFIG dynamics [18].

3. Small signal stability

The small signal stability, and more broadly the dynamic performance, of the power system are related to the damping of the electromechanical modes of oscillation. This oscillatory behaviour is associated fundamentally with (i) the variation in the electrical torque developed by synchronous machines as their rotor angles change; and (ii) the inertia of their rotors. The frequencies associated with these modes of oscillation are typically in the range from 0.5 to 4 Hz.

3.1. Modes of oscillation

The analysis of small signal stability is particularly of interest of electromechanical modes of oscillations. It involves the rotors of individual generators or groups of generators oscillating or swinging against each other. Electromechanical modes of oscillation can be broadly subdivided into (i) local area modes having frequency range of 0.8–2 Hz and (ii) inter-area modes having frequency range of 0.16–0.8 Hz [23,26]. Small signal stability requires that these modes should be adequately damped. The presence of AVR and PSS greatly influence the damping of these modes and can be assessed by means of eigenvalue analysis. The eigenvalue analysis can be carried out by linearizing the system about an operating point and representing it in state space form. For stability, all of the eigenvalues must lie in the left half complex plane. Any eigenvalue in the right half plane denotes an unstable dynamic mode and system instability. The damping contribution provided by any means shifts the location of the eigenvalues associated with the dominant oscillatory modes to the left half of the plane.

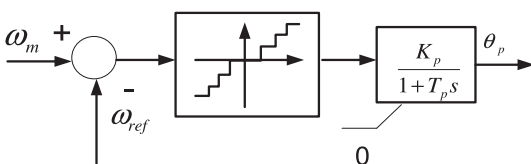


Fig. 2. Pitch angle control scheme.

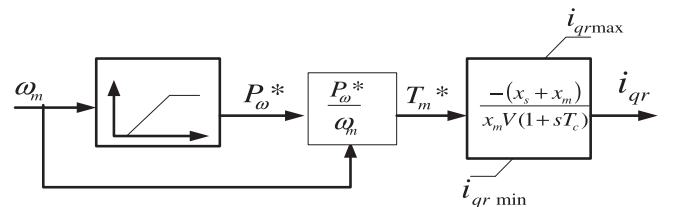


Fig. 3. Rotor speed control scheme.

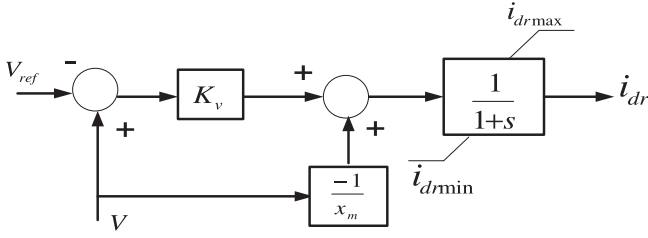


Fig. 4. Voltage control scheme.

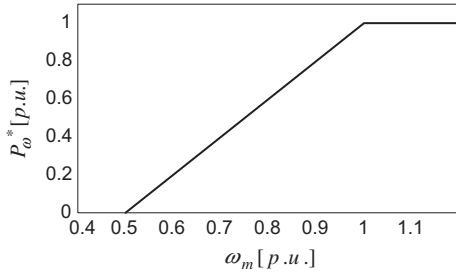


Fig. 5. Power speed characteristics.

3.2. Eigenvalues computation

The behaviour of a dynamical system, such as a power system, can be described by the complete set of n – first order non-linear ordinary differential and algebraic equations (DAE) of the following form [17,23]:

$$\dot{x} = f(x, z, u) \quad (20)$$

$$0 = g(x, z, u) \quad (21)$$

where x takes the values of 1 to n ; n is the order of the system.

At the equilibrium, all time derivatives of the states are zero and are obtained from a load flow analysis.

$$0 = f(x_0, z_0, u_0) \quad (22)$$

$$0 = g(x_0, z_0, u_0) \quad (23)$$

The complete set of differential equations with network equations are arranged in state-space form and linearized about an operating point as:

$$[\Delta \dot{x}] = [A][\Delta x] + [B][\Delta z] \quad (24)$$

$$[0] = [C][\Delta x] + [D][\Delta z] \quad (25)$$

In the small signal perturbed model:

$$[\Delta x] = [\Delta \delta, \Delta \omega, \Delta E]^T \quad (26)$$

$$[\Delta z] = [\Delta \theta_1, \Delta \theta_2, \Delta \theta_3, \Delta V_1, \Delta V_2, \Delta V_3]^T \quad (27)$$

By taking the Laplace transform of above equations

$$\Delta x[s] = \frac{adj(sl - A)}{\det(sl - A)} [\Delta x(0) + B\Delta u(s)] \quad (28)$$

Similarly,

$$\Delta y[s] = C \frac{adj(sl - A)}{\det(sl - A)} [\Delta x(0) + B\Delta u(s)] + D\Delta u(s) \quad (29)$$

The poles of the system are the roots of the characteristic equations given by

$$\det(sl - A) = 0 \quad (30)$$

Above equation can be written as characteristic equation

$$\det(A - \lambda I) = 0 \quad (31)$$

The values of λ ($\lambda = \lambda_1, \lambda_2, \dots, \lambda_n$) which satisfy the characteristic equation, are known as the eigenvalues of matrix A . The numbers of eigenvalues are equal to the number of first-order differential equations considered in the model to represent the system. Eigenvalues may be real or complex, if the matrix is real. The complex eigenvalues always occur in conjugate pairs, as shown

$$\lambda_i = \alpha_i \pm j\beta_i \quad (32)$$

For a given eigenvalue, damping ratio, frequency oscillation, and time constant can be calculated using the following expressions:

$$\zeta = \frac{-\alpha}{\sqrt{\alpha^2 + \beta^2}} \quad f = \frac{\beta}{2\pi} \quad \text{and} \quad t = \frac{1}{\alpha} \quad (33)$$

In this work, eigenvalues have been calculated along with damping ratio and frequency of oscillation for different cases and the participation factor of mode i , can be computed as follows

$$p_i = \begin{pmatrix} p_{1i} \\ p_{2i} \\ \vdots \\ p_{mi} \end{pmatrix} \quad (34)$$

$$p_{ki} = \frac{|\Phi_{ki}| |\Psi_{ki}|}{\sum_{k=1}^n |\Phi_{ki}| |\Psi_{ki}|} \quad (35)$$

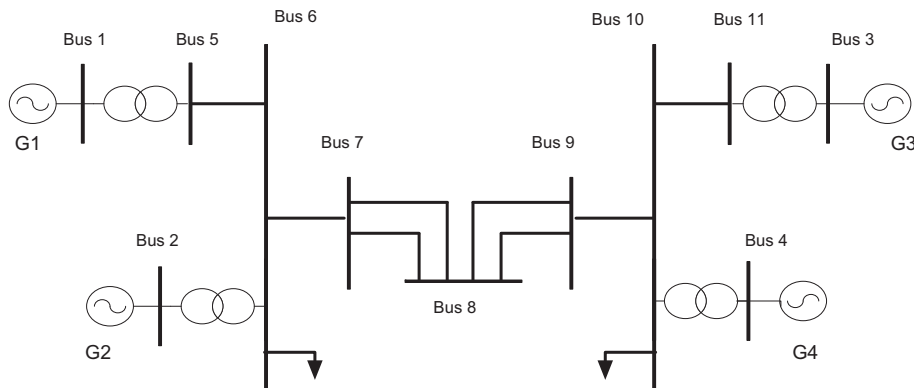


Fig. 6. Two area test system.

Table 1
Identification of states and associated mode numbers.

Cases	Name of associated states	Mode numbers of associated states	Total number of states
Case 1	Synchronous generator $\delta_1, \omega_1, e'_{q1}, e'_{d1}, e''_{q1}, e''_{d1}, \delta_2, \omega_2, e'_{q2}, e'_{d2}, e''_{q2}, e''_{d2}, \delta_3, \omega_3, e'_{q3}, e'_{d3}, e''_{q3}, e''_{d3}, \delta_4, \omega_4, e'_{q4}, e'_{d4}, e''_{q4}, e''_{d4}$	$\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_{22}, \lambda_{23}, \lambda_{24}$	24
Case 2	Synchronous generator $\delta_1, \omega_1, e'_{q1}, e'_{d1}, e''_{q1}, e''_{d1}, \delta_2, \omega_2, e'_{q2}, e'_{d2}, e''_{q2}, e''_{d2}, \delta_3, \omega_3, e'_{q3}, e'_{d3}, e''_{q3}, e''_{d3}, \delta_4, \omega_4, e'_{q4}, e'_{d4}, e''_{q4}, e''_{d4}$ AVR $v_{m1}, v_{r3.1}, v_{f1}, v_{m2}, v_{r3.2}, v_{f2}, v_{m3}, v_{r3.3}, v_{f3}, v_{m4}, v_{r3.4}, v_{f4}$ PSS $v_{1.1}, v_{2.1}, v_{3.1}, v_{1.2}, v_{2.2}, v_{3.2}, v_{1.3}, v_{2.3}, v_{3.3}, v_{1.4}, v_{2.4}, v_{3.4}$	$\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_{22}, \lambda_{23}, \lambda_{24}$ $\lambda_{25}, \lambda_{26}, \dots, \lambda_{35}, \lambda_{36}$ $\lambda_{37}, \lambda_{38}, \dots, \lambda_{47}, \lambda_{48}$	48
Case 3	Synchronous generator $\delta_1, \omega_1, e'_{q1}, e'_{d1}, e''_{q1}, e''_{d1}, \delta_2, \omega_2, e'_{q2}, e'_{d2}, e''_{q2}, e''_{d2}, \delta_3, \omega_3, e'_{q3}, e'_{d3}, e''_{q3}, e''_{d3}$ DFIG $v_{wind}, \theta_p, \omega_m, i_{dr}, i_{qr}$	$\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_{16}, \lambda_{17}, \lambda_{18}$ $\lambda_{19}, \lambda_{20}, \lambda_{21}, \lambda_{22}, \lambda_{23}$	23
Case 4	Synchronous generator $\delta_1, \omega_1, e'_{q1}, e'_{d1}, e''_{q1}, e''_{d1}, \delta_2, \omega_2, e'_{q2}, e'_{d2}, e''_{q2}, e''_{d2}, \delta_3, \omega_3, e'_{q3}, e'_{d3}, e''_{q3}, e''_{d3}$ AVR $v_{m1}, v_{r3.1}, v_{f1}, v_{m2}, v_{r3.2}, v_{f2}, v_{m3}, v_{r3.3}, v_{f3}$ PSS $v_{1.1}, v_{2.1}, v_{3.1}, v_{1.2}, v_{2.2}, v_{3.2}, v_{1.3}, v_{2.3}, v_{3.3}$ DFIG $v_{wind}, \theta_p, \omega_m, i_{dr}, i_{qr}$	$\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_{16}, \lambda_{17}, \lambda_{18}$ $\lambda_{19}, \lambda_{20}, \dots, \lambda_{26}, \lambda_{27}$ $\lambda_{28}, \lambda_{29}, \dots, \lambda_{35}, \lambda_{36}$ $\lambda_{37}, \lambda_{38}, \lambda_{39}, \lambda_{40}, \lambda_{41}$	41
Case 5	Synchronous generator $\delta_1, \omega_1, e'_{q1}, e'_{d1}, e''_{q1}, e''_{d1}, \delta_2, \omega_2, e'_{q2}, e'_{d2}, e''_{q2}, e''_{d2}, \delta_3, \omega_3, e'_{q3}, e'_{d3}, e''_{q3}, e''_{d3}, \delta_4, \omega_4, e'_{q4}, e'_{d4}, e''_{q4}, e''_{d4}$ AVR $v_{m1}, v_{r3.1}, v_{f1}, v_{m2}, v_{r3.2}, v_{f2}, v_{m3}, v_{r3.3}, v_{f3}, v_{m4}, v_{r3.4}, v_{f4}$ PSS $v_{1.1}, v_{2.1}, v_{3.1}, v_{1.2}, v_{2.2}, v_{3.2}, v_{1.3}, v_{2.3}, v_{3.3}, v_{1.4}, v_{2.4}, v_{3.4}$ DFIG $v_{wind}, \theta_p, \omega_m, i_{dr}, i_{qr}$	$\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_{22}, \lambda_{23}, \lambda_{24}$ $\lambda_{25}, \lambda_{26}, \dots, \lambda_{35}, \lambda_{36}$ $\lambda_{37}, \lambda_{38}, \dots, \lambda_{47}, \lambda_{48}$ $\lambda_{49}, \lambda_{50}, \lambda_{51}, \lambda_{52}, \lambda_{53}$	53

m is the number of state variables, p_{ki} is the participation factor of the k th state variable into mode i , Φ_{ki} is the i th element of the k th right eigenvector of A , Ψ_{ki} is the m th element of the i th left eigenvector of A .

4. Proposed approach and test system

Two-area, four machine power system [23] as shown in Fig. 6 has been considered to evaluate the small signal performance in the presence of DFIG. The small signal stability analyses have been carried out for different cases as shown below.

Case 1: All generators are considered as synchronous generators without AVR and PSS.

Case 2: All generators are considered as synchronous generators equipped with AVR and PSS.

Case 3: Any one synchronous generator is replaced by DFIG at a time and remaining synchronous generators are without AVR and PSS.

Case 4: Any one synchronous generator is replaced by DFIG at a time and remaining synchronous generators are equipped with AVR and PSS.

Case 5: Consideration of step wise penetration of wind power by DFIG at bus 2 in the system with remaining generators as synchronous generators equipped with AVR and PSS.

Table 2
Dominant oscillatory modes for Case 1.

Mode number	Eigenvalue	Frequency of oscillation (in Hz)	Damping ratio	Most influential state in the control of the mode with their % participation
λ_9, λ_{10}	$-0.5186 \pm 6.1736j$	0.9860	0.0837	$\delta_1 = 25.6\%, \omega_1 = 25.6\%, \delta_4 = 19.1\%, \omega_4 = 19.1\%$
$\lambda_{11}, \lambda_{12}$	$-0.5096 \pm 5.9979j$	0.9580	0.0846	$\delta_2 = 24.9\%, \omega_2 = 24.9\%, \delta_3 = 19.6\%, \omega_3 = 19.6\%$
$\lambda_{13}, \lambda_{14}$	$-0.1273 \pm 3.1277j$	0.4982	0.0407	$\delta_1 = 14.6\%, \omega_1 = 14.6\%, \delta_4 = 19.1\%, \omega_4 = 9.1\%$

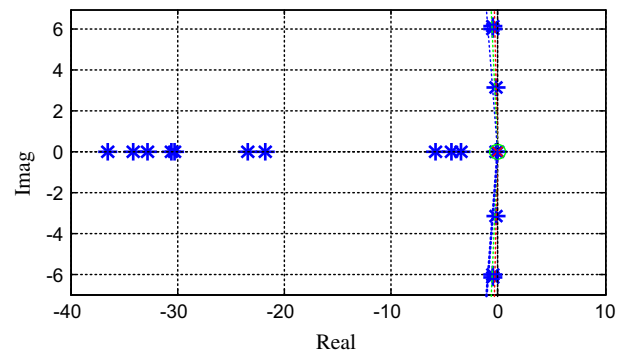


Fig. 7. Plot of eigenvalue for Case 1.

5. Results and discussions

Sixth order synchronous generator, AVR Type III and PSS Type II [25] are considered for the analysis and the values of the parameters are given in Appendix. The name of the associated states and their respective mode numbers for the different cases considered above are listed in Table 1.

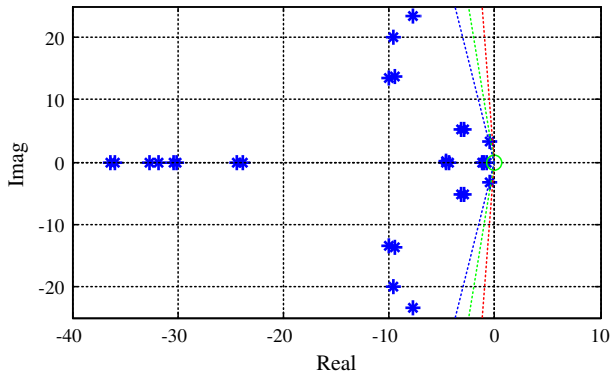


Fig. 8. Plot of eigenvalue for Case 2.

5.1. Case 1: All synchronous generators; without AVR and PSS

The small signal stability analysis has been carried out for the system shown in Fig. 6 for the Case 1 and all 24 eigenvalues are obtained with negative real parts, which indicated that the system is inherently dynamically stable. The plot of obtained eigenvalue is shown in Fig. 7. The system response characteristics are comprised of an amalgam of the dynamic modes identified by system eigenvalues. Hence, it is not possible to directly link specific responses with individual eigenvalues. However, the three dominant oscillatory modes of the test system are shown in Table 2. The observation can be drawn that all eigenvalues have negative real parts, indicating that the system is dynamically stable. The particular network oscillatory response characteristics giving the frequency of oscillations, damping ratio and four most influencing states with their participation are also denoted in Table 2 and classified as follow:

- (1) Mode 1 having a frequency of 0.986 Hz corresponds to the local area mode in Area 1.
- (2) Mode 2 having a frequency of 0.958 Hz corresponds to the local area mode in Area 2.
- (3) Mode 3 having a frequency of 0.498 Hz corresponds to the inter area mode between Area 1 and Area 2.

5.2. Case 2: All synchronous generators; with AVR and PSS

Generator control is exercised by adjustment of the magnitude of the field voltage. The AVR of the excitation control system employs terminal voltage error for the adjustment of the field voltage to control the terminal voltage magnitude. Fast response excitation control system gives fairly good response over the terminal voltage but unfortunately reduces the damping torque of the generator at network oscillation frequencies [27]. This reduction of damping torque can be improved by incorporating a PSS loop into the excitation control system.

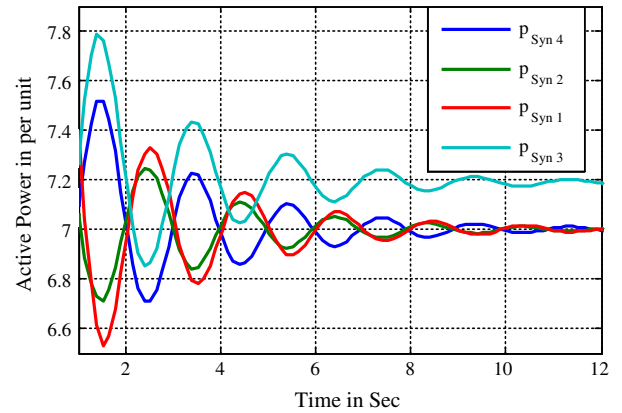


Fig. 9. Transient responses of active power of synchronous generators for Case 2.

For the system shown in Fig. 6, the AVR and PSS have been considered for all four synchronous generators while carrying out the small signal stability analysis. The presence of AVR and PSS changes the number of system states from 24 to 48. The plot of eigenvalues is shown in Fig. 8 for the case under consideration and the network oscillatory response characteristics are also tabulated in Table 3. The results summarized in Table 3 shows that inter-area mode is best damped and frequency of oscillations is also increased with the incorporation of AVR and PSS as compared to the Case 1.

The results of eigenvalue analysis are also confirmed by time domain simulation as shown in Fig. 9. A disturbance of 3-phase fault for the duration of 0.25 s was created at time, $t = 5$ s, at bus 8. For Case 1 and Case 2, the comparative transient responses of rotor speed and angle of synchronous generators are depicted in Figs. 10 and 11, respectively. Though the system is inherently dynamically stable for Case 1, it becomes transiently unstable upon the application for 3 phase fault. On the other hand, the transient oscillations in rotor speed and angle of all synchronous generators are effectively suppressed with the incorporation of AVR and PSS, which also supplement the results obtained by eigenvalue.

5.3. Case 3: Synchronous generator is replaced by DFIG at one bus at a time; remaining synchronous generators are without AVR and PSS

In this case, the synchronous generator at any one bus is replaced by DFIG. The eigenvalue plots for the system after replacing the synchronous generator by DFIG at the different buses are shown in Fig. 12 and results are given in Table 4. The results reveal that the replacement of the synchronous generator by DFIG at any bus makes the system dynamically unstable which can be observed from the inter-area oscillation mode (shown highlighted) which have the positive real part contributing to negative damping ratio. Also, it can be revealed from the results that the frequency of

Table 3
Dominant oscillatory modes for Case 2.

Mode number	Eigenvalue	Frequency of oscillation (in Hz)	Damping ratio	Most influential state in the control of the mode with their % participation
λ_9, λ_{10}	-7.665 ± 23.416	3.9214	0.3110	$e'_{q1} = 16.8\%, e''_{q1} = 12.2\%, e'_{q4} = 12\%, e''_{q4} = 8.6\%$
$\lambda_{11}, \lambda_{12}$	-9.5925 ± 20.0937	3.5437	0.4308	$e'_{q2} = 14.5\%, e'_{q3} = 14.8\%, e'_{q1} = 6.8\%, e'_{q4} = 6.8\%$
$\lambda_{21}, \lambda_{22}$	-9.4783 ± 13.6069	2.6392	0.5715	$e'_{q1} = 16.3\%, e'_{q4} = 19.7\%, e''_{q4} = 10.7\%, \omega_1 = 7.4\%$
$\lambda_{23}, \lambda_{24}$	-9.9886 ± 13.4459	2.6658	0.5963	$e'_{q2} = 16.9\%, e'_{q3} = 19.9\%, e'_{q3} = 10.9\%, \omega_2 = 6.8\%$
$\lambda_{25}, \lambda_{26}$	-3.1904 ± 5.2608	0.97922	0.5185	$\delta_1 = 13\%, \omega_1 = 11.4\%, \delta_4 = 10.6\%, \omega_1 = 9\%$
$\lambda_{27}, \lambda_{28}$	-2.9159 ± 5.2007	0.94893	0.4890	$\delta_3 = 11.6\%, \omega_3 = 10.4\%, \delta_2 = 13.2\%, \omega_2 = 11.5\%$
$\lambda_{29}, \lambda_{30}$	-0.40172 ± 3.2309	0.51817	0.1233	$\delta_4 = 13.8\%, \omega_4 = 12\%, \delta_3 = 12.5\%, \delta_1 = 11.5\%$
$\lambda_{31}, \lambda_{32}$	-4.6668 ± 0.08749	0.74288	0.9998	$e'_{d1} = 30\%, e'_{d2} = 38.9\%, e'_{d3} = 4\%, e'_{d2} = 3.8\%$

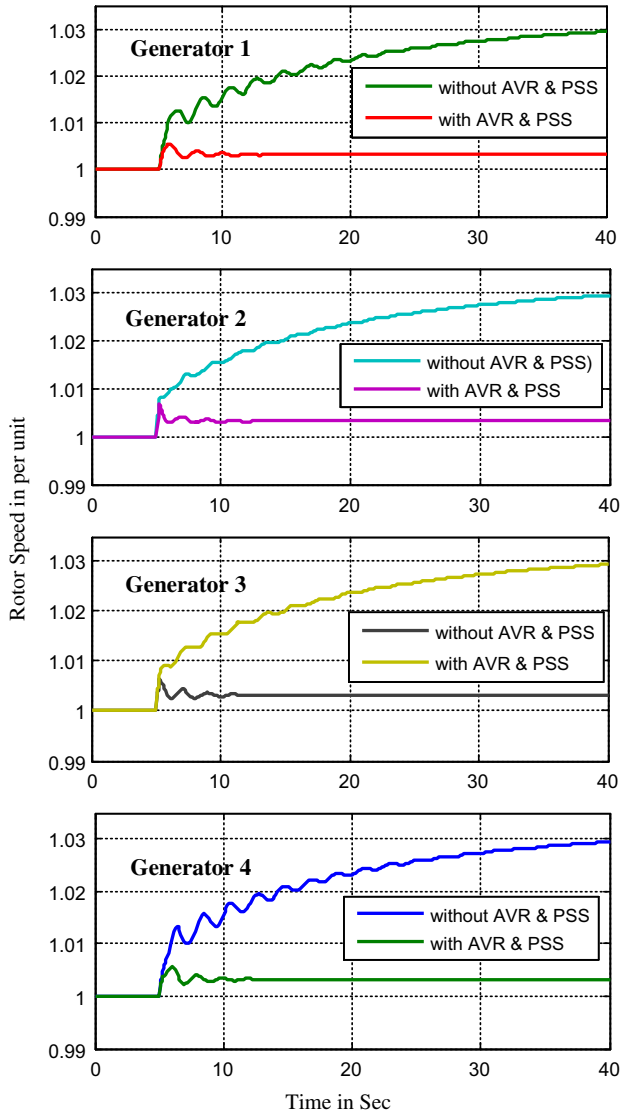


Fig. 10. Comparative transient responses for rotor speed of synchronous generators for Case 1 and Case 2.

oscillation and damping ratio are varying according to the location of DFIG in the system and also responsible for the change in mode shapes. To make the system stable with the large penetration of DFIG at a particular location, the impacts of AVR and PSS at the other synchronous generators in a system are explored in the next section.

5.4. Case 4: Synchronous generator is replaced by DFIG at one bus at a time; remaining synchronous generators are with AVR and PSS

As can be seen from the previous case that the penetration of wind power by DFIG exhibits the intrinsic unstable behaviour, the damping needs to be provided by certain external means such as AVR and PSS located at the synchronous generator. The plot of eigenvalue and results for the case under consideration is shown in Fig. 13 and Table 5, respectively. The results clearly indicate that the effect of AVR and PSS allows the large scale wind power penetration by DFIG without threatening the system stability. It can be seen that placement of DFIG irrespective of location can improve the damping ratio of different modes when operated with AVR and PSS. Hence, the coordinated control of DFIG and AVR, PSS at synchronous generator yields the better damping performance of

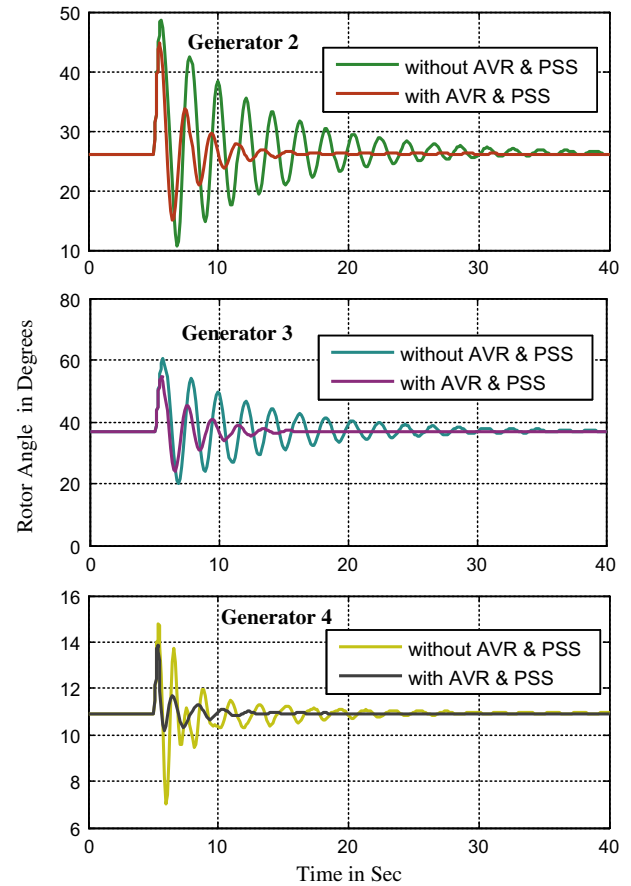


Fig. 11. Comparative transient responses for rotor angle of synchronous generators for Case 1 and Case 2.

local and inter-area modes compared to Case 2. The improvement in damping ratio of the inter-area mode with coordinated control of DFIG and AVR, PSS at synchronous generator is highlighted in Table 5 in comparison to the Case 1 and Case 2 represented in Tables 2 and 3, respectively

5.5. Case 5: Stepwise penetration of wind power by DFIG; Remaining Synchronous Generators are with AVR and PSS

In this case, the influences of stepwise wind power penetration by DFIG at bus 2 on the mode shapes have been analyzed. The wind power penetration in steps of 100 MW is raised with simultaneous reduction in active power contribution by synchronous generator at bus 2. Since the synchronous generator is not fully replaced by DFIG up to the wind power penetration level of 800 MW, the states of both the machines act together and it results in eight oscillatory mode shapes. The variations in oscillatory mode shapes are represented in Fig. 14 with these stepwise wind power penetrations. The dominant Oscillatory Modes for Case 5 with different wind power penetration levels are shown in Table 6. It has been noticed that the mode shapes are not much affected up to certain wind penetration level. Also, the variation in damping ratio of inter-area oscillation mode (mode 8) is slightly improved at higher wind power penetration level. With wind power penetration level in the range of 500 to 800 MW, the variations in local area modes (mode 4 and 5) are ambiguous until synchronous generator is completely replaced by DFIG. Once the synchronous generator at bus 2 is fully replaced by DFIG, the variation in local area modes (mode 3 and 4) show the increment in damping ratios and makes the system

more dynamically stable. It is worthwhile to note that the oscillatory modes and their behaviour with the complete replacement of synchronous generator at bus 2 (wind power penetration of 900 MW) matches with the Case 4 as given in Table 5 when DFIG is located as bus 2.

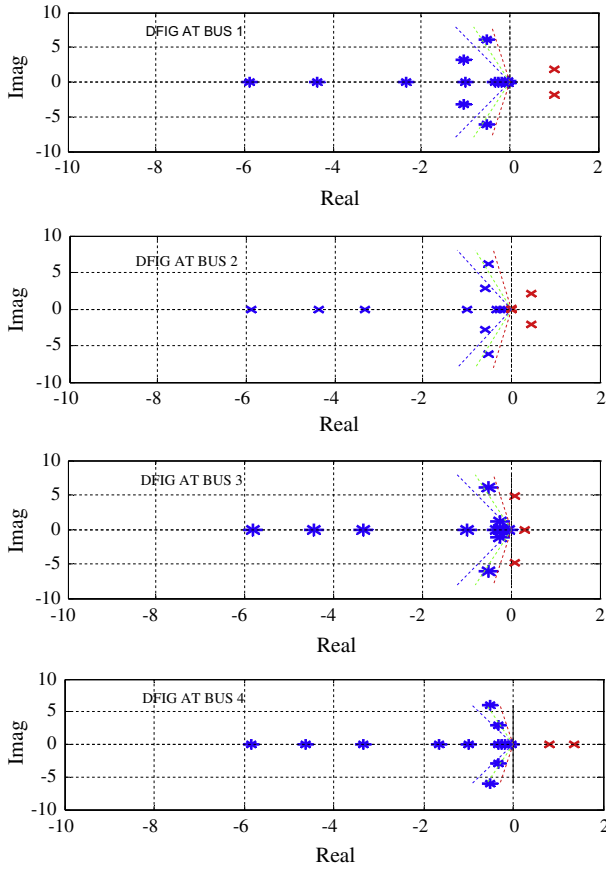


Fig. 12. Plot of eigenvalue for Case 3.

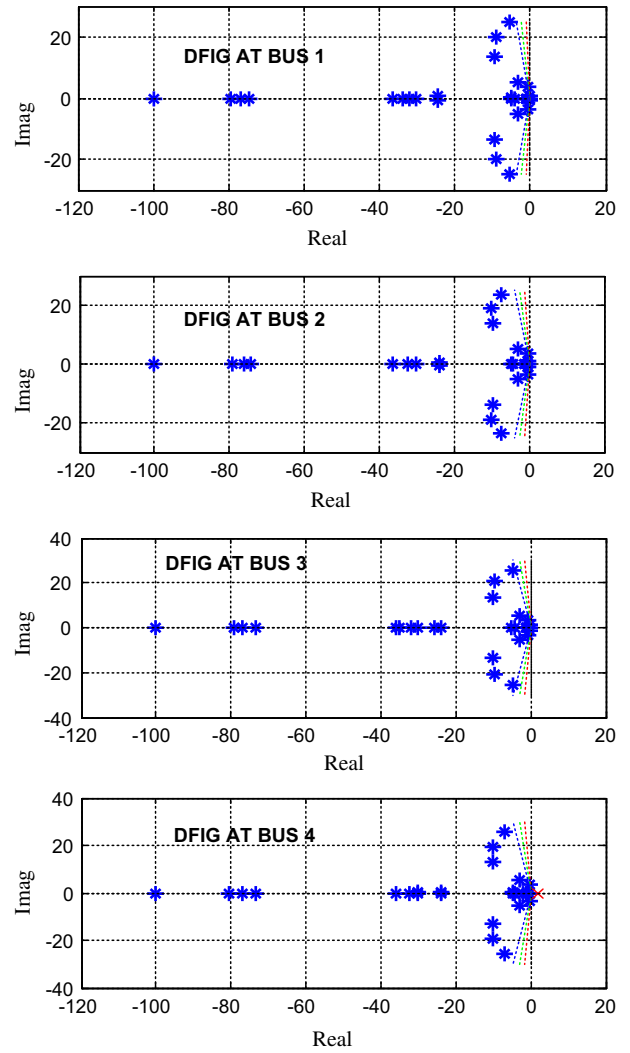


Fig. 13. Plot of eigenvalue for Case 4.

Table 4
Dominant oscillatory modes for Case 3.

Mode number	Eigenvalue	Frequency of oscillation (in Hz)	Damping ratio	Most influential state in the control of the mode with their % participation
<i>Location of DFIG: BUS 1</i>				
λ_{8, λ_9}	-0.52318 ± 6.1613	0.98059	0.0846	$\delta_1 = 25.5\%, \omega_1 = 25.5\%, \delta_3 = 20\%, \omega_3 = 20\%$
$\lambda_{11}, \lambda_{12}$	-1.0411 ± 3.2085	0.51064	0.3086	$\delta_2 = 23.6\%, \omega_2 = 23.6\%, \delta_3 = 8.9\%, \omega_3 = 8.9\%$
$\lambda_{14}, \lambda_{15}$	1.0229 ± 1.7928	0.28533	-0.4955	$\delta_2 = 19\%, \omega_2 = 19\%, \delta_1 = 11.8\%, \omega_1 = 11.8\%$
$\lambda_{18}, \lambda_{19}$	-0.03947 ± 0.0808	0.01287	0.4386	$e'_{q2} = 48.5\%, e'_{q3} = 28.3\%, e'_{q1} = 20.5\%$
<i>Location of DFIG: BUS 2</i>				
λ_{8, λ_9}	-0.51872 ± 6.1612	0.98059	0.0838	$\delta_1 = 25.5\%, \omega_1 = 25.5\%, \delta_3 = 20.2\%, \omega_3 = 20.2\%$
$\lambda_{13}, \lambda_{14}$	-0.60802 ± 2.8666	0.45623	0.2074	$\delta_2 = 23.4\%, \omega_2 = 23.4\%, \delta_3 = 10.5\%, \omega_3 = 10.5\%$
$\lambda_{15}, \lambda_{16}$	0.46071 ± 2.0694	0.32936	-0.21730	$\delta_2 = 21.1\%, \omega_2 = 21.1\%, \delta_3 = 12\%, \omega_3 = 12\%$
$\lambda_{18}, \lambda_{19}$	0.00313 ± 0.0722	0.0115	-0.04328	$e'_{q2} = 49.1\%, e'_{q3} = 27.5\%, e'_{q1} = 21.6\%$
<i>Location of DFIG: BUS 3</i>				
λ_{8, λ_9}	-0.51113 ± 6.0051	0.95574	0.0848	$\delta_2 = 25.6\%, \omega_2 = 25.6\%, \delta_3 = 19.8\%, \omega_3 = 19.8\%$
$\lambda_{10}, \lambda_{11}$	0.1042 ± 4.8121	0.76587	-0.0216	$\delta_1 = 42.8\%, \omega_1 = 42.8\%, e'_{d1} = 3.2\%, e'_{q1} = 2.1\%$
$\lambda_{15}, \lambda_{16}$	-0.24168 ± 1.2105	0.19265	0.1957	$\delta_3 = 22.3\%, \omega_3 = 22.3\%, \delta_2 = 17\%, \omega_2 = 17\%$
λ_{17}	0.2995	0	-1	$e'_{q2} = 39.7\%, e'_{q3} = 32.5\%, e'_{q1} = 7.1\%, \delta_3 = 4.8\%$
<i>Location of DFIG: BUS 4</i>				
λ_{8, λ_9}	-0.51124 ± 5.9989	0.95475	0.0849	$\delta_1 = 25.6\%, \omega_2 = 25.6\%, \delta_2 = 19.8\%, \omega_2 = 19.8\%$
$\lambda_{11}, \lambda_{12}$	-0.32261 ± 2.9975	0.47706	0.1070	$\delta_3 = 25.3\%, \omega_3 = 25.3\%, \delta_2 = 12.9\%, \omega_2 = 12.9\%$
λ_{16}	1.3648	0	-1	$\delta_3 = 15.2\%, \omega_3 = 15.2\%, \delta_1 = 10.7\%, \omega_1 = 10.7\%$
λ_{17}	0.89195	0	-1	$e'_{q1} = 29.6\%, e'_{q2} = 20.3\%, e'_{q3} = 15.8\%, \delta_3 = 4.7\%$

Table 5
Dominant oscillatory modes for Case 4.

Mode number	Eigenvalue	Frequency of oscillation (in Hz)	Damping ratio	Most influential state in the control of the mode with their % participation
<i>Location of DFIG: BUS 1</i>				
λ_8, λ_9	-5.4273 ± 24.9008	4.0561	0.2129	$e'_{q2} = 23.1\%, e''_{q2} = 15.4\%, e'_{q1} = 10.7\%, e''_{q1} = 7.5\%$
$\lambda_{14}, \lambda_{15}$	-8.9891 ± 20.0948	3.5036	0.4083	$e'_{q2} = 17.9\%, e'_{q1} = 12\%, e''_{q2} = 11.8\%, e'_{q3} = 11.8\%$
$\lambda_{16}, \lambda_{17}$	-24.327 ± 0.5506	3.8728	0.9997	$e'_{q2} = 37.2\%, e'_{d3} = 23.8\%, e'_{d1} = 16.5,$
$\lambda_{18}, \lambda_{19}$	-9.4566 ± 13.6167	2.6385	0.5704	$e'_{q3} = 20.4\%, e'_{q1} = 17\%, e''_{q3} = 11.1\%, \omega_1 = 7.7\%$
$\lambda_{20}, \lambda_{21}$	-3.1718 ± 5.2616	0.97779	0.5162	$\delta_1 = 14\%, \omega_1 = 12.4\%, \delta_3 = 11.8\%, \omega_3 = 10\%$
$\lambda_{22}, \lambda_{23}$	-0.5083 ± 3.7749	0.60621	0.1334	$\delta_2 = 27.8\%, \omega_2 = 21.8\%, \delta_3 = 10.4\%, \omega_3 = 9.1\%$
$\lambda_{24}, \lambda_{25}$	-4.8391 ± 0.12474	0.77042	0.9996	$e'_{d2} = 35.7\%, e'_{d2} = 34.2\%, e'_{d2} = 3.7\%, e'_{d2} = 3.6\%$
$\lambda_{27}, \lambda_{28}$	-0.16995 ± 0.7219	0.11805	0.2291	$\delta_2 = 20.3\%, \delta_1 = 16.7\%, \delta_3 = 11.4\%, \omega_2 = 9.7\%$
<i>Location of DFIG: BUS 2</i>				
λ_8, λ_9	-7.2263 ± 23.5663	3.9231	0.2931	$e'_{q1} = 19.9\%, e'_{q1} = 14.2\%, e'_{q3} = 13.8\%, e'_{q3} = 9.8\%$
$\lambda_{11}, \lambda_{12}$	-32.535 ± 0.06085	5.1782	0.9999	$v_{2.1} = 36.7\%, v_{2.3} = 18.5\%, v_{2.2} = 16.6\%$
$\lambda_{14}, \lambda_{15}$	-9.9679 ± 19.0282	3.4188	0.4640	$e'_{q2} = 33.6\%, e'_{q2} = 22.7\%, v_{m.1} = 10.4\%$
$\lambda_{16}, \lambda_{17}$	-24.063 ± 0.49938	3.8306	0.9997	$e'_{d2} = 41.5\%, e'_{d3} = 24.7\%, e'_{d1} = 18\%,$
$\lambda_{18}, \lambda_{19}$	-9.474 ± 13.6384	2.6429	0.5705	$e'_{q3} = 20.8\%, e'_{q1} = 16.2\%, e'_{q3} = 11.4\%, \omega_1 = 7.6\%$
$\lambda_{20}, \lambda_{21}$	-3.1534 ± 5.2577	0.97575	0.5143	$\delta_1 = 13.8\%, \omega_1 = 12.3\%, \delta_3 = 12.1\%, \omega_3 = 10.1\%$
$\lambda_{22}, \lambda_{23}$	-0.38984 ± 3.6198	0.57943	0.1070	$\delta_2 = 30.1\%, \omega_2 = 25\%, \delta_3 = 8.3\%, \omega_3 = 7.5\%$
$\lambda_{24}, \lambda_{25}$	-4.6942 ± 0.14419	0.74746	0.9995	$e'_{d2} = 41\%, e'_{d1} = 35.5\%, e'_{d2} = 4.1\%, e'_{d3} = 5.6\%$
$\lambda_{27}, \lambda_{28}$	-0.26422 ± 1.2167	0.19815	0.2122	$\delta_1 = 18.3\%, \delta_2 = 15.2\%, \delta_3 = 14.6\%, \omega_3 = 11.5\%$
<i>Location of DFIG: BUS 3</i>				
λ_8, λ_9	-4.7252 ± 25.272	4.0919	0.1837	$e'_{q1} = 34.3\%, e'_{q1} = 22.7\%, v_{m.3} = 10.9\%, e'_{d1} = 5.5\%$
$\lambda_{10}, \lambda_{11}$	-9.6733 ± 20.5351	3.6127	0.4261	$e'_{q2} = 19.2\%, e'_{q3} = 18.9\%, e'_{q2} = 13.6\%, e'_{q3} = 13.2\%$
$\lambda_{18}, \lambda_{19}$	-10.211 ± 13.2716	2.6651	0.6098	$e'_{q3} = 19.7\%, e'_{q2} = 18.5\%, e'_{q3} = 10.9\%, e'_{q3} = 10.1\%$
$\lambda_{20}, \lambda_{21}$	-2.9401 ± 5.2183	0.95327	0.4908	$\delta_2 = 14.6\%, \omega_2 = 12.8\%, \delta_3 = 12.2\%, \omega_3 = 10.9\%$
$\lambda_{22}, \lambda_{23}$	-0.867 ± 3.6021	0.58967	0.2340	$\delta_1 = 32.4\%, \omega_1 = 25.4\%, \delta_3 = 7.2\%, \omega_3 = 6\%$
$\lambda_{24}, \lambda_{25}$	-0.23608 ± 1.0483	0.17102	0.2197	$\delta_2 = 17.5\%, \delta_1 = 16.4\%, \delta_3 = 15.1\%, \omega_3 = 12.4\%$
$\lambda_{30}, \lambda_{31}$	-4.4559 ± 0.04145	0.7092	0.9999	$e'_{d3} = 36.1\%, e'_{d2} = 33.2\%, e'_{q3} = 6\%, e'_{d3} = 3.3\%$
<i>Location of DFIG: BUS 4</i>				
λ_8, λ_9	-7.1295 ± 25.8042	4.2607	0.2663	$e'_{q1} = 23.6\%, e'_{q1} = 18.3\%, e'_{q2} = 11.6\%, v_{m.3} = 10.2\%$
$\lambda_{11}, \lambda_{12}$	-10.253 ± 19.4692	3.5021	0.4659	$e'_{q1} = 18.1\%, e'_{q3} = 13.7\%, e'_{q1} = 12.9\%, e'_{q2} = 11.4\%$
$\lambda_{14}, \lambda_{15}$	-30.2187 ± 0.2111	4.8096	0.9999	$v_{2.3} = 30.4\%, e'_{d2} = 21\%, e'_{d3} = 17.9\%, v_{2.1} = 9.8\%$
$\lambda_{16}, \lambda_{17}$	-24.1284 ± 0.4887	3.8409	0.9997	$e'_{d1} = 41\%, e'_{d3} = 17.5\%, e'_{d2} = 17.5\%, e'_{d1} = 4.1\%,$
$\lambda_{18}, \lambda_{19}$	-10.235 ± 13.2901	2.6698	0.6101	$e'_{q3} = 19.9\%, e'_{q2} = 18\%, e'_{q3} = 11.1\%, e'_{q2} = 9.9\%$
$\lambda_{20}, \lambda_{21}$	-2.9229 ± 5.2173	0.95179	0.4887	$\delta_2 = 14.2\%, \omega_2 = 12.8\%, \delta_3 = 12.4\%, \omega_3 = 11.1\%$
$\lambda_{22}, \lambda_{23}$	-0.31873 ± 3.4892	0.55763	0.09096	$\delta_1 = 25.6\%, \omega_1 = 24.5\%, \delta_3 = 12.5\%, \omega_3 = 10.6\%$
$\lambda_{25}, \lambda_{26}$	-4.6361 ± 0.22359	0.73872	0.9988	$e'_{d2} = 38.4\%, e'_{d1} = 29.4\%, e'_{d3} = 9.3\%, e'_{q2} = 2\%$

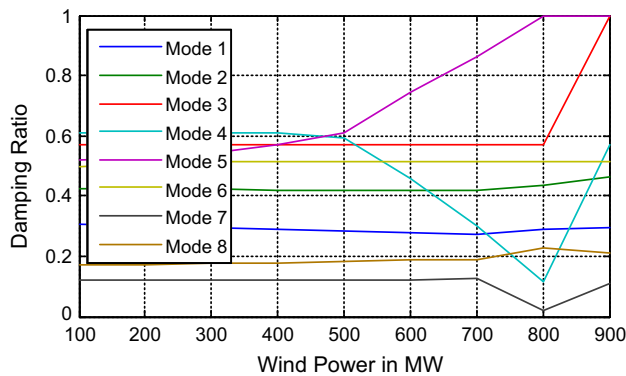


Fig. 14. Wind power vs. damping ratio for Case 5.

6. Conclusion

The penetration of DFIG based wind power generation may affect the small signal stability of the system while displacing the synchronous generation in order to accommodate wind generation. In this paper, the comprehensive assessment of small signal stability of two-area interconnected power system has been presented with and without DFIG. The presence of DFIG can alter

the local and inter-area mode shapes and shows the dynamically unstable behaviour, when the additional damping torques provided from AVR and PSS of the synchronous generators are not available. However, the system operating with DFIG penetration along with AVR and PSS on synchronous generator can have the beneficial effect on dynamic performance and improves the damping ratio of local and inter-area modes irrespective of its location. The effect of wind power penetration cannot affect the mode shapes in larger extent up to the certain level which can become ambiguous for the higher levels of penetration.

Appendix A

Synchronous generator

Machine dynamic model = 6th order; power rating = 900 MVA; voltage rating = 20 kV.

Frequency rating = 60 Hz; armature resistance $r_a = 0.025$ pu; leakage reactance $x_l = 0.2$ pu.

d -axis synchronous reactance $x_d = 1.8$ pu; d -axis transient reactance $x'_d = 0.3$ pu.

d -axis sub-transient reactance $x''_d = 0.25$ pu.

d -axis open circuit transient time constant $T'_{d0} = 8$ s.

d -axis open circuit sub-transient time constant $T''_{d0} = 0.03$ s.

Table 6

Dominant oscillatory modes for Case 5 with different wind power penetration levels.

Mode number	Eigenvalue	Frequency of oscillation (in Hz)	Damping ratio	Most influential state in the control of the mode with their % participation
<i>Penetration level of wind power by DFIG = 100 MW</i>				
Mode 1	-7.5878 ± 23.6537	3.9535	0.3055	$e'_{q1} = 15.5\%, e''_{q1} = 11.3\%, e'_{q4} = 11\%, e'_{q2} = 9.8\%$
Mode 2	-9.5139 ± -20.2346	3.5586	0.4255	$e'_{q3} = 14\%, e'_{q2} = 13.1\%, e''_{q3} = 9.6\%, e''_{q2} = 9.1\%$
Mode 3	-9.48 ± 13.6132	2.6402	0.5715	$e'_{q4} = 20\%, e'_{q1} = 16.4\%, e'_{q4} = 10.9\%, e''_{q1} = 8.8\%$
Mode 4	-10.038 ± 13.1422	2.632	0.6070	$e'_{q3} = 19.3\%, e'_{q2} = 17.6\%, e'_{q3} = 10.5\%, e'_{q2} = 9.5\%$
Mode 5	-3.2192 ± 5.2898	0.9855	0.5199	$\delta_1 = 10.3\%, \omega_1 = 8.9\%, \delta_4 = 7.9\%, \omega_4 = 6.9\%$
Mode 6	-3.0249 ± 5.3009	0.9713	0.4956	$\delta_2 = 10.9\%, \omega_2 = 9.2\%, \delta_3 = 8.1\%, \omega_3 = 7.5\%$
Mode 7	-0.40917 ± -3.3697	0.5402	0.1205	$\delta_3 = 16.1\%, \omega_3 = 14\%, \delta_4 = 11.7\%, \omega_4 = 10.3\%$
Mode 8	-0.14605 ± -0.8495	0.1371	0.1694	$\delta_1 = 16\%, \delta_4 = 11.8\%, \delta_2 = 10.7\%, \delta_3 = 10.3\%$
<i>Penetration level of wind power by DFIG = 500 MW</i>				
Mode 1	-7.0827 ± 23.9415	3.9736	0.2837	$e'_{q1} = 14.8\%, e''_{q1} = 10.7\%, e'_{q4} = 10.2\%, e'_{q2} = 9.6\%$
Mode 2	-9.2309 ± -20.0058	3.5066	0.4190	$e'_{q3} = 18.4\%, e''_{q3} = 12.4\%, e'_{q2} = 7.1\%, e'_{q1} = 8.3\%$
Mode 3	-9.478 ± -13.6257	2.6416	0.5710	$e'_{q4} = 20.7\%, e'_{q1} = 16.6\%, e'_{q4} = 11.3\%, e''_{q1} = 8.9\%$
Mode 4	-7.7532 ± 10.4944	2.0766	0.5942	$e'_{q2} = 21.9\%, v_{2,1} = 16\%, e'_{q3} = 11.2\%, \omega_2 = 12.6\%$
Mode 5	-5.0608 ± -6.6254	1.3269	0.6070	$e'_{q2} = 17.6\%, \delta_2 = 11.9\%, v_{2,1} = 16.8\%, \omega_2 = 10.5\%$
Mode 6	-3.1508 ± 5.2692	0.97711	0.5132	$\delta_1 = 13.5\%, \delta_4 = 12.2\%, \omega_1 = 12.1\%, \omega_4 = 10.2\%$
Mode 7	-0.43579 ± -3.6195	0.58022	0.1195	$\delta_3 = 23.5\%, \omega_3 = 20.8\%, \delta_4 = 9.1\%, \omega_4 = 8\%$
Mode 8	-0.19994 ± 1.0795	0.17473	0.1821	$\delta_1 = 17.8\%, \delta_4 = 14.1\%, \delta_1 = 10.9\%, \omega_4 = 11\%$
<i>Penetration level of wind power by DFIG = 700 MW</i>				
Mode 1	-6.8759 ± 24.3582	4.0282	0.2717	$e'_{q1} = 13.9\%, e''_{q1} = 10.1\%, e'_{q4} = 9.5\%, e'_{q3} = 12.9\%$
Mode 2	-9.202 ± -20.0934	3.5174	0.4164	$e'_{q3} = 19.8\%, e''_{q3} = 13.3\%, e'_{q1} = 9.2\%, e'_{q4} = 8.8\%$
Mode 3	-9.4772 ± -13.6253	2.6415	0.5710	$e'_{q4} = 20.7\%, e'_{q1} = 16.6\%, e'_{q4} = 11.3\%, e''_{q1} = 8.9\%$
Mode 4	-3.5526 ± 11.1876	1.8682	0.3027	$e'_{q2} = 19.7\%, v_{2,1} = 17.2\%, \delta_2 = 11.2\%, \omega_2 = 23.2\%$
Mode 5	-8.5823 ± -5.0854	1.5877	0.8603	$e'_{q2} = 23.7\%, v_{2,1} = 25.6\%, e'_{q3} = 8.9\%, \omega_2 = 8.2\%$
Mode 6	-3.1538 ± 5.2694	0.97738	0.5136	$\delta_1 = 13.6\%, \delta_4 = 12.2\%, \omega_1 = 12.2\%, \omega_4 = 10.2\%$
Mode 7	-0.46653 ± -3.7231	0.59718	0.1243	$\delta_3 = 26.4\%, \omega_3 = 23.6\%, \delta_4 = 8\%, \omega_4 = 7\%$
Mode 8	-0.21599 ± 1.1206	0.18163	0.1893	$\delta_1 = 18.5\%, \delta_4 = 14.8\%, \omega_1 = 11.6\%, \omega_4 = 11.7\%$
<i>Penetration level of wind power by DFIG = 800 MW</i>				
Mode 1	-7.2154 ± 23.9436	3.98	0.2885	$e'_{q1} = 17\%, e''_{q1} = 12.3\%, e'_{q4} = 8.8\%, e'_{q3} = 12.7\%$
Mode 2	-9.4866 ± -19.7125	3.4817	0.4336	$e'_{q3} = 28.5\%, e''_{q3} = 19.2\%, v_{m,2} = 8.9\%, e'_{q4} = 6.7\%$
Mode 3	-9.4756 ± -13.6269	2.6416	0.5709	$e'_{q4} = 20.7\%, e'_{q1} = 16.7\%, e'_{q4} = 11.3\%, e''_{q1} = 9\%$
Mode 4	-0.84955 ± 7.2165	1.1565	0.1169	$\delta_2 = 31.4\%, \omega_2 = 31.4\%, \delta_3 = 8.4\%, \omega_3 = 8.4\%$
Mode 5	-7.389	–	1	$e'_{d2} = 87.4\%, e'_{d2} = 11.5\%$
Mode 6	-3.1461 ± -5.2612	0.97562	0.5132	$\delta_1 = 13.8\%, \delta_4 = 12.1\%, \omega_1 = 12.7\%, \omega_4 = 10.1\%$
Mode 7	-0.05857 ± 3.3665	0.53588	0.0174	$\delta_3 = 18.1\%, \omega_3 = 13.4\%, \delta_4 = 9.3\%, \omega_4 = 8.6\%$
Mode 8	-0.28867 ± 1.233	0.20155	0.2280	$\delta_1 = 15.6\%, \delta_4 = 12.2\%, \delta_3 = 16.8\%, \omega_3 = 11.8\%$

q -axis synchronous reactance $x_q = 1.7$ pu; q -axis transient reactance $x'_q = 0.55$ pu.

q -axis sub-transient reactance $x''_q = 0.25$ pu.

q -axis open circuit transient time constants $T'_{q0} = 0.4$ s.

q -axis open circuit sub-transient time constant $T''_{d0} = 0.05$ s.

Mechanical starting time ($2 \times$ Inertia constant) ($M = 2H$) = 13 kW s/kV A (for G1 and G2).

($M = 2H$) = 12.35 kW s/kV A (for G3 and G4), damping co-efficient = 0 pu.

d -axis additional circuit leakage time constant $T_{AA} = 0.002$ s.

Transformer

Power rating = 900 MVA; voltage ratio = 20 kV/230 kV; resistance = 0 pu; reactance = 0.15 pu.

Fault

Type = 3 phase fault; power rating = 100 MV A; voltage rating = 400 kV.

Frequency = 60 Hz; fault time = 5 s; fault clearing time = 0.25 s.

Fault resistance = 0.001 pu; fault reactance = 0.001 pu.

Automatic voltage regulator

Type = 3; maximum regulator gain = 7 pu; minimum regulator gain = -6.4 pu.

Regulator gain = 200; regulator pole $T_2 = 1$ s; regulator zero $T_1 = 1$ s.

Time constant of field circuit $T_d = 0.001$ s.

Time delay of the measurement system $T_r = 0.015$ s.

Coefficient of ceiling function, $A = 0.0006$, $B = 0.9$.

Power system stabilizer

PSS model = 2; input signal = omega; maximum stabilizer output = 0.2 pu.

Minimum stabilizer output = -0.2 pu; gain $k_w = 9.5$; time constant $T_w = 1.41$ s.

Electrical power gain $K_p = 20$; electrical voltage gain $K_v = 5$.

Lead lag time constants $T_1 = 0.154$ s, $T_2 = 0.033$ s, $T_3 = 1$ s, $T_4 = 1$ s.

Gain $K_a = 25$; time constant $T_a = 0.5$.

Additional signal limits $V_{a-max} = 0.045$, $V_{a-thr} = 0.045$, $V_{3-max} = 0.045$, $V_{3-min} = -0.045$.

Threshold limits $E_{-thr} = 1$, $W_{-thr} = 0.95$ pu.

Doubly fed induction generator

Power rating = (2 MV A * 450); voltage rating = 20 kV; frequency = 60 Hz.

Stator resistance $R_s = 0.01$ pu; stator reactance $X_s = 0.10$ pu; rotor resistance $R_r = 0.01$ pu.

Rotor reactance $X_r = 0.08$ pu; magnetization reactance $X_m = 3$ pu.

Inertia constants, $H_m = 3$ kW s/kV A; pitch control gain = 10 pu; time constants = 3 s.

Voltage control gain $K_v = 10$ pu; power control time constant $T_e = 0.01$ s.

Number of poles = 4; gear box ratio = 1:89; blade length = 75.0 m; number of blade = 3.

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